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Unmanned Wingless Rotorcraft Scaling

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The work addresses the allometric scaling of unmanned rotorcraft. Compound unmanned rotorcraft were not considered in this study. A population of 93 unmanned rotorcraft is compiled into a single dataset comprising size and performance parameters, including maximum take-off weight, maximum payload weight, battery mass, battery capacity, rotor diameter, endurance, range, maximum speed, and operational ceiling. Rotorcraft are grouped based on rotor configuration, propulsion type, and perceived application to study the clustering of the data points. An apparent performance limit for unmanned electric rotorcraft is found at 120 minutes endurance and 70 kilometers range; higher range and endurance are currently only achievable using thermal propulsion systems. Further, the regression analysis links size and performance parameters in easy-to-use expressions for the preliminary design of unmanned rotorcraft. It is determined that size parameters have a stronger correlation with maximum take-off weight than performance parameters do. Comparative analysis suggests that unmanned rotorcraft cannot be treated as a sub-class of conventional helicopters due to distinct scaling behaviors. A comparison with theoretical models based on first-order physics principles highlights the major role of low-Reynolds number aerodynamic effects in rotorcraft scaling.

Nomenclature

A	=	net actuation area; as defined in Eq. (2), m^2
D	=	rotor diameter, m
E_b	=	battery energy capacity, Wh
g	=	gravitational acceleration, $m\,s^{-2}$
H	=	ceiling altitude, m
I_k	=	power to kinetic energy conversion coefficient, s
I_p	=	power to potential energy conversion coefficient, s
k	=	generic coefficient of proportionality

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KE = kinetic energy; $\frac{1}{2}mV^2$, J

m = generic mass, kg

m_b = battery mass, kg

N_P = number of rotors

P = rotor power, W

P_e = battery discharge power, W

P_{max} = maximum available power, W

P_{shaft} = rotor shaft power, W

P_{ideal} = ideal induced power in hover at sea level; as defined in Eq. (3), W

P_r = maximum available power to shaft power ratio; as defined in Eq. (6)

PE = potential energy; mgH, J

R^2 = coefficient of determination

Re = Reynolds number

S = range, km

t = maximum endurance time, min

T = rotor thrust, N

V = maximum speed, $\text{km}\,\text{h}^{-1}$

W = maximum take-off weight, kg

W_p = maximum payload weight, kg

ρ = density, $\text{kg}\,\text{m}^{-3}$

κ = power correction factor

κ_e = discharge to ideal induced power ratio; P_e/P_{ideal}

$\tilde{\mu}_3$ = moment coefficient of skewness

σ = standard deviation

I. Introduction

The popularity of the "camera drone" (unmanned rotary-wing system for aerial photography and videography) made the unmanned rotorcraft into a mass-produced consumer product. This success story, as well as technological advancements in electronics components miniaturization, batteries, and autopilots, sparked the interest of industry and academia to explore other applications where unmanned rotorcraft can be used. Consequently, there is now an abundance of commercial off-the-shelf unmanned rotorcraft spanning several orders of magnitude in size, but their scaling behavior has not been studied sufficiently. This work addresses the allometric scaling of unmanned rotorcraft.

The analysis is based on a population of commercially available rotorcraft and four prototypes of cargo wingless multirotors for urban air mobility. The objective is to determine the regression between rotorcraft primary performance and size parameters and contrast these with conventional rotorcraft data.

A similar paper by Lorenz [1] aimed to describe the allometric relations for fixed-wing, rotary-wing, and lighter-than-air vehicles. Lorenz compiled data on 64 rotorcraft, with a maximum take-off weight (MTOW) from 0.1 kg to 50 000 kg, and only ten of these had a mass less than 500 kg. Lorenz collected aircraft MTOW, rotor radius, installed power, and maximum speed. Hence, found empirical relations to describe the correlation between these. Other work in this field was accomplished by Filippone [2] and Rand and Khromov [3], who used statistical means to describe populations of manned rotorcraft. Although there have been follow-up studies on fixed-wing aircraft such as [4], no studies looked at rotary-wing aircraft in the past two decades. Moreover, there has been a significant increase in the number of commercial off-the-shelf unmanned rotorcraft since the 2010s, driven by the commercialization and affordability of "camera drones". This study extends the allometric analysis to smaller rotorcraft scales and introduces new categories, such as endurance, range, battery mass, battery capacity, rotor configuration, and propulsion type. The allometric relations would be useful for preliminary design analyses, as well as studying the effect of the scale of unmanned aerial systems, e.g., for drone-delivery networks or environmental monitoring.

A well-known study by Stolaroff et al. [5] assessed the effect of rotorcraft scale on the specific energy consumption of drone delivery networks in support of comparison with conventional road-based transportation networks. They used the momentum theory to model the power of two aircraft, a tetra-rotor and an octa-rotor, with "representative" maximum payload masses of 0.5 kg and 8.1 kg, respectively. The study found that the tetra-rotor was seemingly more suitable for drone-delivery operations and therefore recommended that drone size should be kept limited as much as possible. However, no evidence was presented as to why the particular values that emerged from their momentum theory model were actually representative of tetra- and octa-rotor vehicles. For example, the payload represented 19.5% of the tetra-rotor base-case take-off weight and approximately 29.2% for the octa-rotor, respectively. Also, the base-case disk loading was 12.5 kg m^{-2} for the tetra-rotor and the 20.5 kg m^{-2} for the octa-rotor, respectively. Therefore, this raises the question: are the conclusions of this well-known study due to the incongruities mentioned above, or are smaller rotary-wing aircraft actually more effective than larger ones?

Typically, rotorcraft sizing studies couple aerodynamic power modeling (e.g., using momentum theory) with individual component scaling along the powertrain, which is decomposed into a battery, electronic speed controller, motor, and rotor [6–16]. The coupled system takes inputs such as flight time, payload mass, and rotor radius and iterates until the system's MTOW converges, giving a powertrain weight breakdown. The methodology is then verified against known aircraft configurations to estimate modeling error. The main problem with these types of studies is the inconsistencies between the assumed scaling laws. For instance, Bershadsky et al. [8] assumed that the airframe mass represents 19% of MTOW based on a population of a few aircraft, while Budinger et al. [12] assumed a proportion of

29%. This makes comparisons between studies difficult and raises questions about the veracity of the conclusions. By contrast to typical rotorcraft studies, this paper aims to describe how rotorcraft size influences the main performance parameters based on a statistically-significant population of unmanned rotorcraft. By extending the analysis to an unmanned rotorcraft dataset and comparing it with conventional helicopters, scaling behavior associated with small-scale phenomena may be revealed, the importance of which might have been neglected or poorly understood before. This study also introduces energy-based criteria that divide the combined conventional and unmanned rotorcraft population into clusters based on propulsion technology and rotorcraft class.

II. Methodology

A. Data Collection

A conventional helicopter database was obtained from [2] and complemented with additional categories from [17]. Data on unmanned rotorcraft were collected from multiple sources [18–49], including rotorcraft manufacturers' online catalogs and the Vertical Flight Society (VFS) electric vertical take-off and landing (eVTOL) directory [50]. No unmanned compound rotorcraft were present in this study. The most prominent manufacturers of unmanned rotorcraft were identified based on a web search; the online product catalogs were itemized and appended to the database according to rotorcraft specifications sheets. In general, different manufacturers reported different performance parameters. For example, some manufacturers reported maximum endurance at a given advance ratio [18], while others did not report the flight regime in which the maximum endurance was achieved [34]. Where no datum was available, the entry was left blank, increasing the database sparsity. For marketing purposes, some parameters may be exaggerated under the assumption of ideal flight conditions (such as air temperature, wind speed, and operating altitude) and may not reflect the actual rotorcraft performance. In the absence of a consensus within the industrial and academic rotorcraft community, some database sparsity could not be avoided, which has also been flagged by Zhang et al. [51].

Because the database did not feature custom-built aircraft, such as those found in academic research projects, it is expected that some variation in design performance parameters could have been lost. The data was limited to *unmanned* rotorcraft since conventional helicopters are a more mature technology; instead, where applicable, this paper compares the present work on unmanned rotorcraft with established analyses on rotorcraft allometry, such as described in Chapter 6 in [52] and in [1–3].

B. Data Grouping

The dataset comprised 93 unmanned rotorcraft described by 12 parameters. The parameters encapsulate information about the classification, performance, and size of the rotorcraft, as defined below. Table 1 and Table 2 offer a statistical summary of the performance and size parameters, respectively.

1. Classification Parameters

Rotor type: describes the rotor configuration of an aircraft, e.g., "O1", "C4". The first character specifies the rotor type: "C" for coaxial, "D" for ducted, and "O" for open rotors. The second character corresponds to the number of rotors in the propulsion group. There were 93 non-empty rotor-type entries in the dataset. The modal rotor type was "O4", i.e., open tetra-rotor configuration (47 entries). The least frequent rotor types were coaxial hexa-rotor, coaxial octa-rotor, and open 16-rotor configuration (1 entry each). The distribution of unmanned rotorcraft based on the rotor type is shown in Fig. 1a.

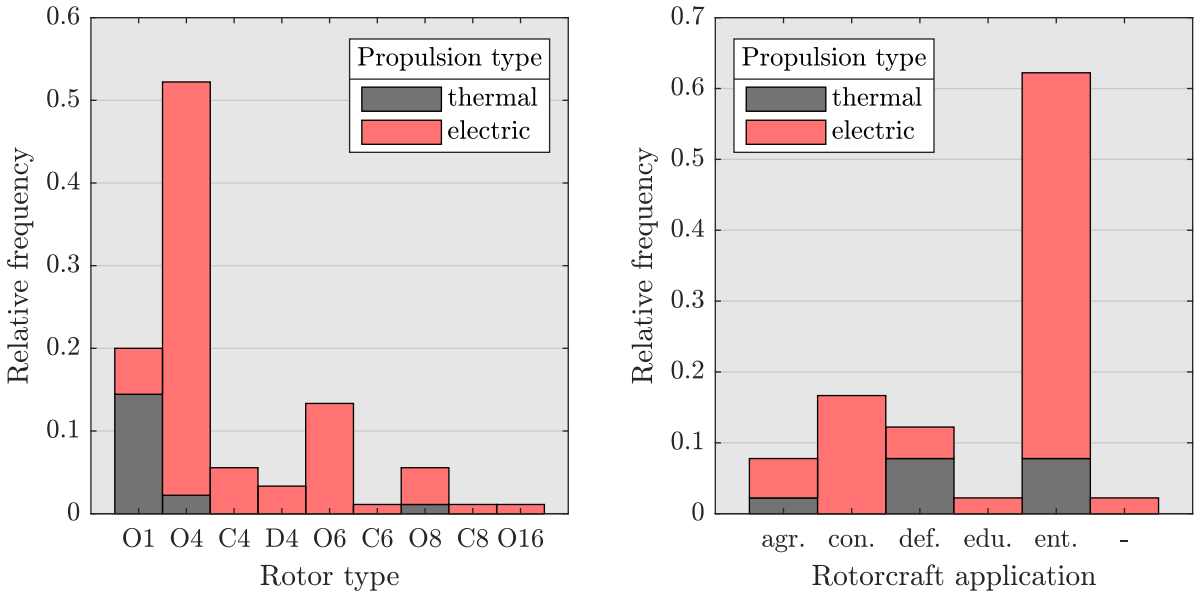
Application: describes the perceived use of a rotorcraft. Five subcategories were identified: agricultural, consumer, defense, educational, and enterprise. Only a fraction of rotorcraft manufacturers directly stated the intended use case. If the use case was not present, the authors manually assigned an application entry based on manufacturer scope, the most apparent use scenario, and vehicle acquisition costs. Commercial and educational rotorcraft were on the lower end of the MTOW spectrum, while the defense, agricultural, and enterprise rotorcraft were on the upper end. The most common application was enterprise (56 entries), the least common was educational (2 entries), and there were 91 non-empty entries. Figure 1b shows the distribution of the rotorcraft based on the type of perceived application.

Propulsion type: indicates whether the rotorcraft were electrically or thermally powered. The propulsion type indicated the form of energy storage. Electrical rotorcraft used batteries, and thermal used fossil fuels as the primary energy source. To clarify, this work classified hybrid-propulsion rotorcraft, i.e., which used fossil fuels to power electric motors, as thermal propulsion rotorcraft. Neither gas turbines nor hydrogen fuel-cell technology were present in the collected dataset. The propulsion-type entries were fully populated in the dataset. The database featured 16 thermal propulsion vehicles and 77 electric. Thermal engines only appeared above the 14 kg mark, which is likely due to a technological limit on the feasible miniaturization of that technology.

2. Performance Parameters

Endurance: represents the maximum flight time in minutes [min], subject to the size of fuel tanks or battery and the payload configuration. Technical specifications appeared inconsistent across manufacturers, who did not usually provide a thorough description of the advance ratio and payload fraction at which endurance was obtained. In this work, the endurance was interpreted as the hover endurance at zero payload. The minimum endurance was 7.2 min, the maximum was 480 min, and the median was 40.2 min. There were 91 non-empty entries.

Range: maximum distance in kilometers [km] that can be covered by the aircraft. Similar to endurance, the operating condition at which the range was measured is not usually described. This parameter was interpreted as the maximum ferry range of the rotorcraft. Some manufacturers may have reported the range limited by the data link distance, which is a performance limitation of such a class of vehicles [25]. Therefore, this parameter does not consistently reflect the performance of rotorcraft as in the helicopter nomenclature definition.The minimum was 1 km, the maximum was



(a) Rotor type grouping **(b) Application grouping**

Fig. 1 Unmanned rotorcraft classification.

1020 km, the median was 21 km. There were 49 non-empty entries.

Maximum speed: in kilometers per hour [$\mathrm{km\,h^{-1}}$]. Some manufacturers reported the maximum speed limited using software [18]. Nonetheless, this number was interpreted as the maximum speed as in its conventional definition. The minimum value of maximum speed was 22 $\mathrm{km\,h^{-1}}$, the maximum was 240 $\mathrm{km\,h^{-1}}$, and the median was 70 $\mathrm{km\,h^{-1}}$. There were 80 non-empty entries.

Ceiling: represents the maximum operational altitude, provided in meters [m] above sea level. Some manufacturers specify a maximum tested ceiling, and others claim a theoretical operating ceiling. This parameter is interpreted as the maximum hover altitude at MTOW. The minimum ceiling was 30 m, the maximum was 7000 m, the median was 4500 m. There were 68 non-empty entries.

Table 1 Performance parameters summary.

Parameter	Notation	Unit	Min	Median	Max	Mean	σ[a]	$\tilde{\mu}_3$[b]	Entries
Endurance	t	min	7.2	40.2	480	82.2	105.1	2.11	91
Range	S	km	1	21	1020	79	190	3.91	49
Maximum speed	V	$\mathrm{km\,h^{-1}}$	22	70	240	78	42	1.65	80
Ceiling	H	m	30	4500	7000	4200	1760	−0.54	68

[a] σ - standard deviation
[b] $\tilde{\mu}_3$ - moment coefficient of skewness

3. Sizing Parameters

Maximum take-off weight: represents the maximum take-off weight in kilograms [kg]. The MTOW was used as the independent size parameter for the analysis because it was the most consistently reported size parameter by the manufacturers. The minimum MTOW was 0.03 kg, the maximum was 800 kg, and the median was 12 kg. There were 91 non-empty entries.

Maximum payload: represents the maximum payload capacity of a vehicle in kilograms. For fully-electric rotorcraft, maximum payload is usually specified for the smallest available battery (lowest-capacity battery). Increased sparsity in data can be observed in the 0.1 kg to 1 kg MTOW range because, at that mass range, consumer rotorcraft with camera payloads dominate. For "camera drones" with fixed payloads, the maximum payload weight was omitted by the manufacturers. This is likely because some aerial photography/videography rotorcraft such as DJI™ Mavic 3 [18] have a fixed payload that is embedded in the structure of the aircraft, and further payload cannot be accommodated. The minimum value for maximum payload was 0.015 kg, the maximum was 225 kg, and the median was 9.5 kg. The number of non-empty entries was 60.

Rotor diameter: represents the diameter in meters of an individual rotor. For multirotor aircraft, the size of the rotors was identical. For helicopters, the main rotor diameter was used. The rotor diameter ranged 0.045 m to 7.7 m. The median rotor diameter was 0.46 m. The number of non-empty entries was 69.

Battery mass: the total mass of the batteries given in kilograms. Battery mass was only collected for electric rotorcraft. The battery mass ranged from 0.01 kg to 12 kg. The median was 0.37 kg. The number of non-empty entries was 34.

Battery capacity: the total energy capacity of the batteries expressed in watt-hours [Wh]. Battery capacity was only collected for electric rotorcraft. The minimum battery capacity was 0.74 Wh, the maximum was 4560 Wh, the median was 89 Wh. The number of non-empty entries was 50.

Table 2 Size parameters summary.

Parameter	Notation	Unit	Min	Median	Max	Mean	σ[a]	$\tilde{\mu}_3$[b]	Entries
MTOW	W	kg	0.03	11	800	68	161	3.14	91
Maximum payload	W_p	kg	0.02	9.5	225	30	55	2.59	60
Rotor diameter	D	m	0.05	0.46	7.7	1.2	1.7	2.32	69
Battery mass	m_b	kg	0.01	0.37	12	1.6	2.9	2.38	34
Battery capacity	E_b	Wh	0.74	89	4560	390	720	4.12	50

[a] σ - standard deviation
[b] $\tilde{\mu}_3$ - moment coefficient of skewness

C. Analysis Procedure

The regression analysis was performed in the MATLAB® environment using the MathWorks® native Statistics and Machine Learning Toolbox. The primary goal of the analysis was to explore the data via clustering and regression techniques and to extract easy-to-use empirical relations. Following that, the objective was to contrast the empirical models with those derived from perfect allometric scaling principles, which could be considered as working hypotheses.

Perfect allometric scaling means that all volume-based properties of an object would scale with length to the power of three, i.e., $V \propto L^3$, all area-based properties with length to the power of two, $A \propto L^2$, and mass would scale with volume as follows, $M \propto V$. In this case, volume-based properties translate to aerodynamic properties, e.g., related to viscous effects and turbulent mixing. Assuming geometric scaling and constant rotorcraft system density, the total actuation area, and the aircraft weight are expected to follow the square-cube law; therefore, the first hypothesis is

$$A \propto W^{2/3} \tag{1}$$

where W is the maximum take-off weight and A is the net actuation area, given as follows

$$A = \pi \left(\frac{D}{2}\right)^2 N_P \tag{2}$$

where D is an individual rotor diameter and N_P is the number of rotors in the rotorcraft configuration, assuming all rotors in a multirotor system are identical. The total actuation area, A, represents the sum of all primary rotor areas in the rotor system; this definition is applicable for both single- and multi-rotor aircraft. According to the momentum theory, the ideal power in hover at maximum take-off weight is

$$P_{\text{ideal}} = \frac{T^{3/2}}{(2\rho A)^{1/2}} \tag{3}$$

where ρ is the ambient air density in [kg m^{-3}] and T is the rotor thrust corresponding to the maximum take-off weight. Assuming the first hypothesis is correct, as in [15], and substituting Eq. (1) in Eq. (3), it can be demonstrated that the rotor power should scale according to

$$P \propto \frac{W^{3/2}}{A^{1/2}} \propto \frac{W^{3/2}}{(W^{2/3})^{1/2}} = W^{7/6} \tag{4}$$

For electric aircraft operating at maximum endurance conditions, the battery discharge power is

$$P_e = \frac{E_b}{t} \tag{5}$$

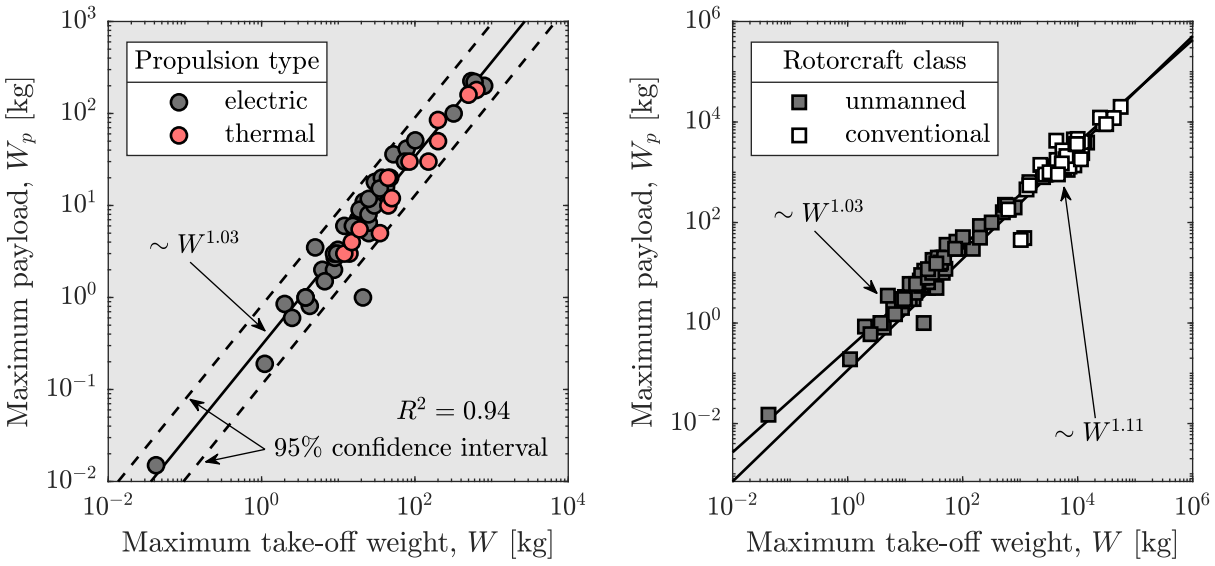
where E_b is the battery energy in watt-hours and t is the maximum endurance time as provided by the manufacturer,

given in hours. Considering perfect allometric scaling, the second hypothesis is that the P_e and P_{ideal} scales isometrically. For an electric rotorcraft, the endurance is a function of battery energy capacity and the rate of energy dissipation $t = E_b/P$. Assuming constant battery-specific energy and battery mass fraction, it is expected that $t \propto W/P$. Further, assuming ideal power scaling, $P \propto W^{7/6}$, the third hypothesis states that the endurance for electric rotorcraft scales according to $t \propto W^{-1/6}$, i.e., the endurance shallowly reduces with rotorcraft size.

III. Results

A. Sizing Parameters

Figure 2a shows maximum payload mass variation with MTOW according to $W_p = 0.30 \cdot W^{1.03}$, also provided for convenience in Table 3. This represents approximately isometric scaling, meaning that the ratio between the maximum payload mass and MTOW is constant across the entire rotorcraft range, irrespective of propulsion and rotor type. This is in agreement with the data on the conventional rotorcraft class [2], shown in Fig. 2b. This allometric relation can be used during the preliminary phases of unmanned rotorcraft design to quickly estimate MTOW based on a given maximum payload mass.



(a) Propulsion type grouping **(b) Rotorcraft class comparison**

Fig. 2 Maximum payload vs maximum take-off weight of a rotorcraft population.

The regression analysis in Fig. 3 shows that battery mass scales with MTOW according to $m_b = 0.29 \cdot W^{0.87}$. The relation suggests that, for larger rotorcraft, the battery mass represents a relatively smaller proportion of MTOW than predicted by perfectly allometric scaling. A possible explanation for this behavior is that larger electrical components may be more efficient than their miniaturized versions, therefore reducing the necessary battery energy [7, 53–55].

Speculatively, it could also be that large batteries do not scale well due to internal resistance build-up and heat dissipation effects making large batteries untenable choices. Not plotted here, but the relation between battery mass and battery capacity was $E_b = 177 \cdot m_b^{0.98}$, with a determination coefficient $R^2 = 0.98$, based on the unmanned electric rotorcraft population. This means that battery capacity scales almost perfectly allometrically with battery mass [7, 8, 10].

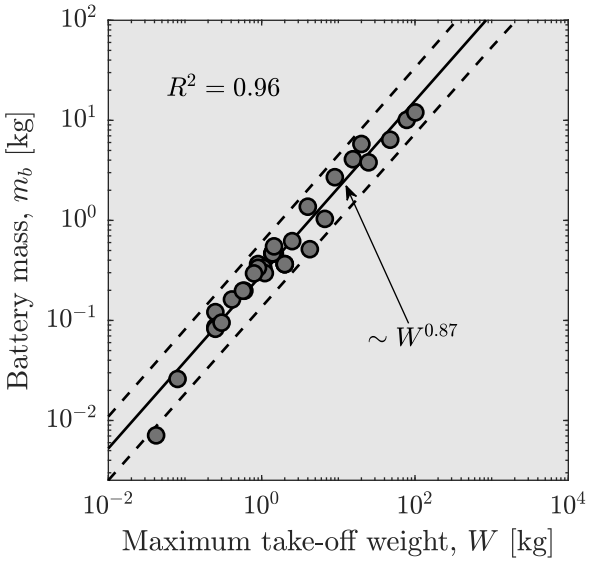


Fig. 3 Battery mass vs maximum take-off weight.

Figure 4a shows the total actuation area scales proportionally to MTOW according to $A = 0.13 \cdot W^{0.87}$. The square-cube law predicts that $A \propto W^{2/3}$; this means that the rotor area exhibits positive allometric scaling. The three aircraft outside the 95% confidence interval are ducted tetra-rotors, as annotated in Fig. 4b, which typically have a higher disk loading. The comparison with conventional rotorcraft, shown in Fig. 4c, suggested that conventional rotorcraft actually have a negative allometric scaling. This is consistent with the data in Chapter 6 in [52], which attributed this effect to nonaerodynamic constraints. In reality, it is observed that for conventional rotorcraft, the number of blades and rotor solidity increase with vehicle size [2], which explains the negative allometry of the area with MTOW. On the other hand, the analysis by Lorenz [1] found that, for conventional rotorcraft, the rotor area exhibits positive allometry, $A \propto W^{0.80}$, which contradicts the findings and discussion in [2, 52].

From Fig. 4c, it can be seen that the total actuation area of unmanned rotorcraft is lower than predicted by extrapolation of the regression analysis of the conventional helicopter population. This can be attributed to the fact that at a small scale, the maximum available power to shaft power ratio is disproportionately large for the size of the rotorcraft [56]. Additionally, this could also be due to the multirotor form factor and volumetric constraints that affect multirotor more than single-rotor aircraft. Moreover, mission profiles within the visual line of sight, typical for unmanned rotorcraft, may involve more acceleration and deceleration due to reduced operational volume, which could

also be a cause for unmanned rotorcraft to be overpowered. This explains an apparently steeper gradient of the area regression curve for the unmanned population compared with the conventional population.

Figure 4d shows the disk loading variation with MTOW for unmanned and conventional rotorcraft. It can be seen that electric rotorcraft exhibit lower disk loadings than conventional rotorcraft. This could be due to less stringent structural constraints at smaller scales. Moreover, low Reynolds number effects at a smaller scale increase the effective disk loading [52]. Therefore, the disk loading may be smaller by design to compensate for this performance penalty, which is explained in more detail in Sec. III.B. Although conventional rotorcraft do not suffer from low Reynolds number effects, structural and material constraints are a bigger concern.

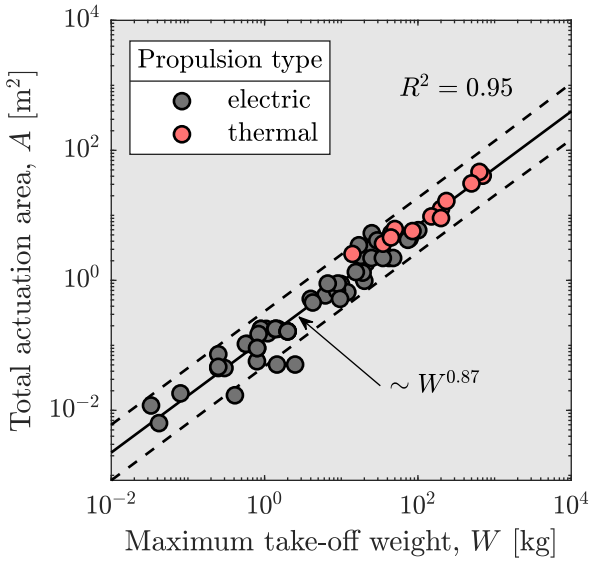
B. Performance Parameters

There was no noticeable pattern between MTOW and maximum speed, as evidenced by a low coefficient of determination, $R^2 = 0.23$. This finding is consistent with the analysis by Lorenz [1], who attributed this phenomenon to the high complexity of helicopters. In the case of unmanned rotorcraft, software limitations might result in an extra layer of uncertainty and inconsistency in the data. Similarly, there was no correlation between MTOW and ceiling. However, incorporating the maximum speed and the ceiling into kinetic and potential energies, respectively, helped to discriminate between conventional, thermal, and electric unmanned rotorcraft, as shown in Fig. 5. This analysis is similar to that by [1], who used maximum speed and maximum installed power in the form of $k \cdot W^x \cdot V^y$. The maximum installed power was unavailable, but the maximum potential and kinetic energy were used instead, as the maximum ceiling and the maximum velocity can be linked to the maximum available power.

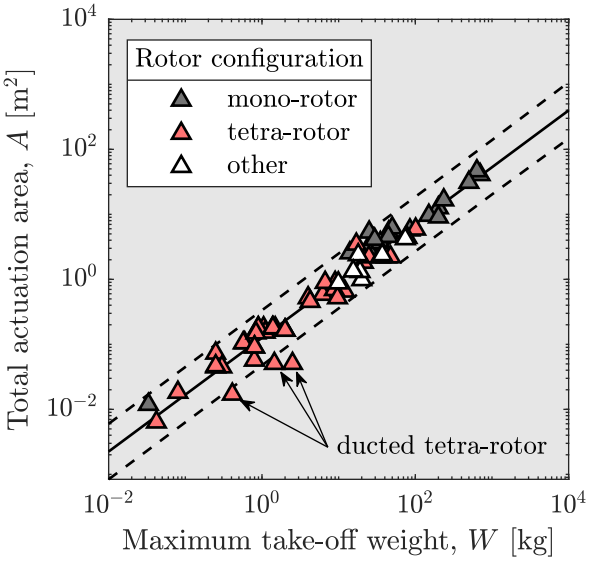
Figure 5c shows the ratio of the kinetic energy to ideal induced power. This ratio incorporates non-ideal aerodynamics effects, κ, the capabilities of the system to convert power into maintaining the state of maximum kinetic energy, I_k, and the ratio of the maximum available power to the shaft power, P_r. The ratio of kinetic energy to ideal induced power in hover can thus be decomposed as

$$\frac{KE}{P_{\text{ideal}}} = \frac{I_k P_{\max}}{P_{\text{ideal}}} = I_k \left(\frac{P_{\max}}{P_{\text{shaft}}}\right)\left(\frac{P_{\text{shaft}}}{P_{\text{ideal}}}\right) = I_k P_r \kappa \tag{6}$$

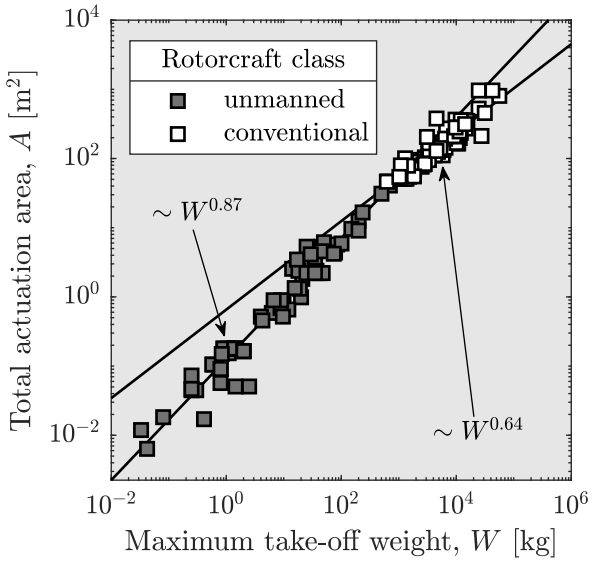
where $P_{\max}$ is the maximum available power; P_{shaft} and P_{ideal} are the shaft power and the ideal power at sea-level conditions, respectively. The results suggested that unmanned aircraft have a lower kinetic energy to ideal power ratio than conventional rotorcraft. This could be attributed to parasitic power playing a more important role for smaller rotorcraft and the fact that multirotors rely on pitching the entire airframe for thrust vectoring as they lack swashplate mechanisms. Furthermore, software-based limitations on maximum speeds could also affect multirotor aircraft. At the same time, for larger aircraft, the ratio is limited by the tip Mach number, which results in constant kinetic energy to ideal power ratio.



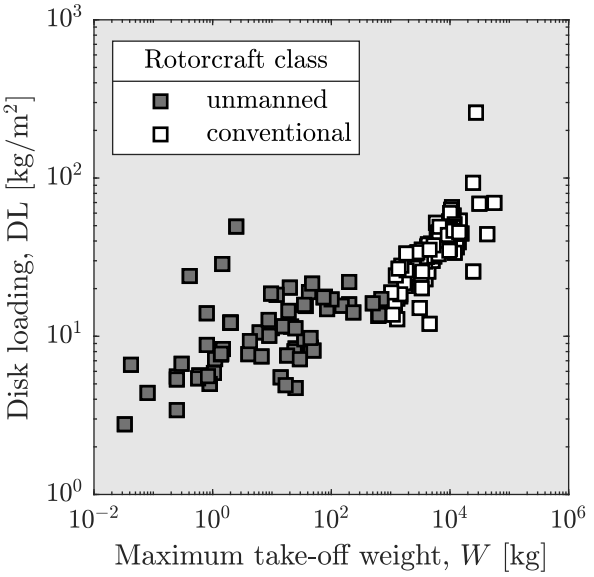
(a) Unmanned rotorcraft propulsion type grouping



(b) Unmanned rotorcraft rotor configuration grouping



(c) Rotorcraft class comparison



(d) Disk loading comparison

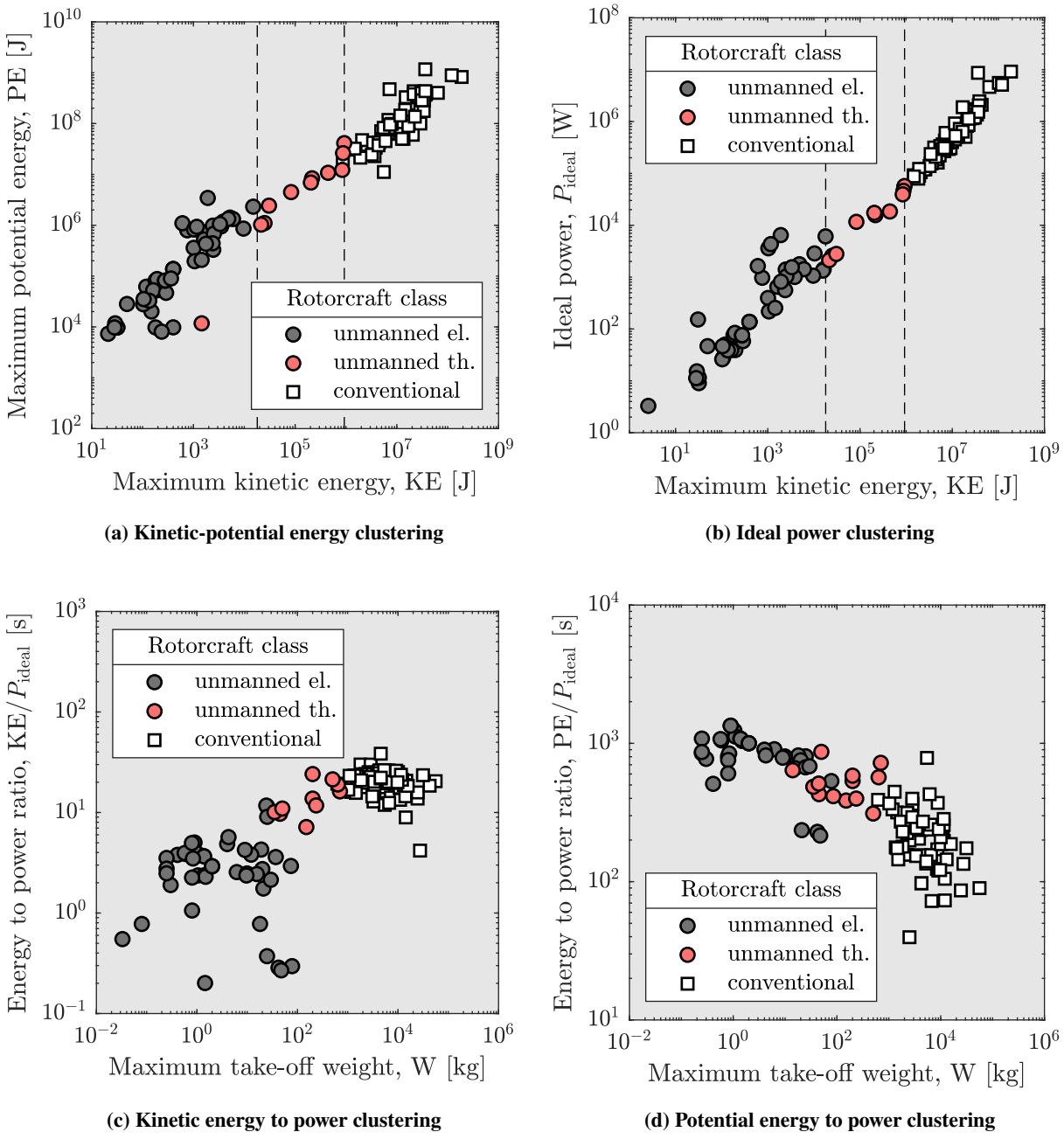
Fig. 4 Net actuation area of a rotorcraft vs maximum take-off weight.

Figure 5d shows the ratio of the potential energy to ideal induced power, which is analogously decomposed as

$$\frac{PE}{P_{\text{ideal}}} = I_p P_r \kappa \tag{7}$$

where I_p is a measure of the capability of the system to convert power into maintaining the state of maximum potential energy. Figure 5d shows that the potential energy to power ratio is smaller for conventional rotorcraft compared to

unmanned rotorcraft. This could be because smaller rotorcraft have relatively more excess power than their larger counterparts.



(a) Kinetic-potential energy clustering

(b) Ideal power clustering

(c) Kinetic energy to power clustering

(d) Potential energy to power clustering

Fig. 5 Maximum potential and kinetic energy. Note, *el.* and *th.* stand for electric and thermal, respectively.

According to the results in Fig. 6a, the ideal power regression shows a good agreement with perfect allometric scaling. However, the battery discharge power scales with MTOW according to $P_e = 114 \cdot W^{0.79}$, as shown in Fig. 6b. This exhibits negative allometry since the second hypothesis predicted $P \propto W^{7/6}$. Simultaneously, the ratio between

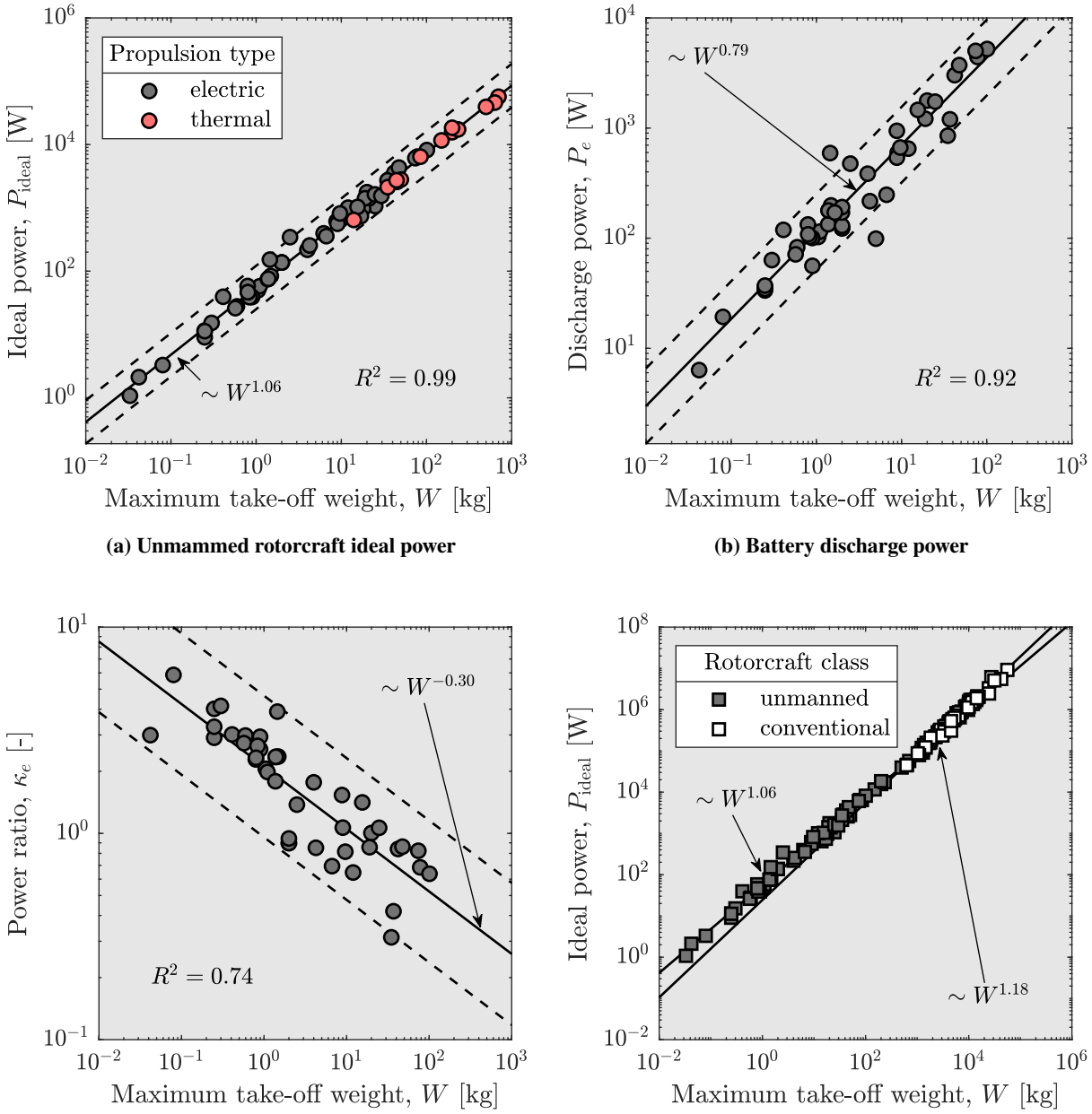
the discharge and the ideal power, κ_e, is inversely proportional to MTOW, as depicted in Fig. 6c; this matches the results in [1]. The power ratio is a measure of the electric rotorcraft's overall efficiency. It incorporates both losses in the electrical powertrain and non-ideal aerodynamics effects such as induced and viscous losses. The power ratio is intended to bridge between required battery power and ideal induced power. Small-scale rotorcraft appear to consume more power than predicted by momentum theory and geometric scaling. The power penalty is likely caused by low Reynolds number aerodynamic effects and powertrain efficiency degradation [1, 7, 52–55, 57–59]. Another possible reason for the power penalty is the low quality of rotor blades at small scales [60]. Although ideal power shows a good agreement between manned and conventional rotorcraft, as shown in Fig. 6d, it is not a good metric for comparing the two rotorcraft classes. Lorenz [1] found that the installed power of conventional helicopters scales with MTOW as $P \propto W^{1.10}$, which is closer to perfect allometric scaling. This disparity suggests that unmanned rotorcraft cannot simply be considered as a sub-class of full-scale conventional helicopters due to aerodynamic and technological factors.

Based on Fig. 7a, which shows the flight endurance against MTOW, two clusters are distinguishable, corresponding to electric and thermal propulsion types. Note that the endurance was not usually measured at the MTOW reported by the manufacturer. For electric rotorcraft, the best-fit relation between endurance and MTOW is $t = 0.49 \cdot W^{0.09}$, which is in contrast with the third hypothesis, $t \propto W^{-1/6}$. This outcome is unexpected, also considering that the battery mass had a negative allometric scaling with respect to MTOW, $m_b = 0.29 \cdot W^{0.87}$. No comparison could be made with conventional helicopters because the corresponding data on endurance does not exist in the literature. Figure 7b, unmanned rotorcraft range, similarly shows two clusters for electric and thermal propulsion. The transition from electric to thermal propulsion takes place at around 70 km range and 120 min of endurance. After this, thermal propulsion is superior to electric propulsion by approximately one order of magnitude in terms of range and endurance. This further evidences the difference between electric and thermal technology, highlighting the potential limitations of wingless multirotor for urban air mobility, given the current technological level.

Range and endurance affect the suitability of an unmanned rotorcraft for a given application, yet during data collection, it was found that these parameters were not usually reported coherently. Moreover, even when reported, confounding factors persisted; these included unclear assumptions (e.g., safety factors, depth-of-discharge), terminology (e.g., maximum flight time and maximum hover time), and testing conditions (e.g., atmospheric conditions, repeatability) [51]. From a transparency perspective, it would be beneficial for the academic and industrial communities to "develop a consensus" on measuring and reporting unmanned rotorcraft performance [51].

IV. Conclusions

An allometric study was conducted on the size and performance parameters of an unmanned rotorcraft population with maximum take-off weight ranging from 0.03 kg to 800 kg. The rotorcraft population was manually gathered from different sources and summarized based on 12 parameters, describing rotorcraft size and performance. The regression



(a) Unmammed rotorcraft ideal power

(b) Battery discharge power

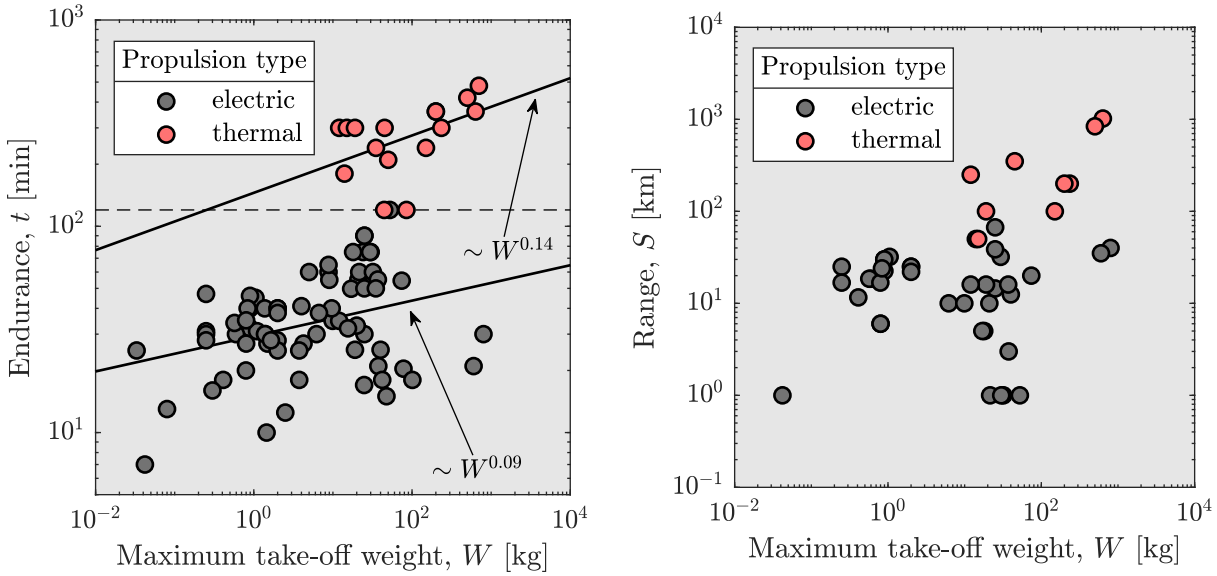
(c) Battery discharge to ideal power

(d) Rotorcraft class comparison

Fig. 6 Power vs maximum take-off weight.

study was carried out, and scaling laws were established. Further, the unmanned rotorcraft population was compared against conventional helicopters over fundamental size-performance relations.

The rotorcraft exhibited a strong correlation between the maximum take-off weight and size parameters (battery mass, battery capacity, net actuation area, and maximum payload weight). The regression between maximum take-off weight and performance parameters (maximum endurance, maximum range, maximum flight speed, and operational



(a) Endurance **(b) Range**

Fig. 7 Mission performance parameters.

Table 3 Regression equations.

Parameter	Equation	R^2	MPE[a]	MAPE[b]	RMSPE[c]	Rotorcraft population
Maximum payload	$W_p = 0.30 \cdot W^{1.03}$	0.94	0.12	0.38	0.83	unmanned
Maximum payload	$W_p = 0.12 \cdot W^{1.11}$	0.80	0.22	0.52	1.12	conventional [2]
Battery mass	$m_b = 0.29 \cdot W^{0.87}$	0.96	0.06	0.28	0.40	electric unmanned
Battery capacity	$E_b = 177 \cdot m_b^{0.98}$	0.98	0.03	0.19	0.30	electric unmanned
Battery capacity	$E_b = 53 \cdot W^{0.85}$	0.93	0.11	0.39	0.67	electric unmanned
Total actuation area	$A = 0.13 \cdot W^{0.87}$	0.95	0.13	0.40	0.77	unmanned
Total actuation area	$A = 0.65 \cdot W^{0.64}$	0.83	0.03	0.19	0.26	conventional [2]
Ideal power	$P_{\text{ideal}} = 55 \cdot W^{1.06}$	0.99	0.02	0.16	0.22	unmanned
Ideal power	$P_{\text{ideal}} = 24 \cdot W^{1.18}$	0.99	0.01	0.10	0.14	conventional [2]
Discharge power	$P_e = 114 \cdot W^{0.79}$	0.92	0.10	0.34	0.59	electric unmanned
Power ratio	$\kappa_e = 2.11 \cdot W^{-0.30}$	0.74	0.07	0.30	0.42	electric unmanned

[a] MPE - mean percentage error
[b] MAPE - mean absolute percentage error
[c] RMSPE - root mean squared percentage error

ceiling) was weaker; this can be attributed to inconsistencies in the performance reported by unmanned rotorcraft manufacturers and a lack of standardized testing methods.

The comparative analysis based on combined size and performance criteria revealed that unmanned rotorcraft exhibit distinct allometric behavior to conventional helicopters. Hence, regression relations derived for conventional helicopters cannot be applied to unmanned rotorcraft, suggesting that unmanned rotorcraft cannot be treated as a scaled

sub-class of conventional helicopters. The regression analysis indicated that the unmanned rotorcraft performance degrades as the aircraft size decreases due to aerodynamic effects associated with low Reynolds numbers and poor powertrain efficiency at small rotorcraft scales. The range and endurance performance of thermal propulsion rotorcraft are superior to electrical propulsion rotorcraft. The transition between electric and thermal propulsion takes place at around 70 km range and 120 min of endurance. This reveals a technological limit that must be addressed for large-scale electric rotorcraft to become a competitive type of aerial vehicle.

The presented data and analysis are useful for rotorcraft engineers and aerial roboticists who aim to study the effects of aircraft scale on a given problem, such as urban air mobility, environmental monitoring, or drone delivery services. Future work should focus on developing novel battery technologies to fill the gap between the electric and thermal propulsion rotorcraft. Small-scale aerodynamics should be investigated to develop mitigation strategies for the apparent performance limitation of small rotorcraft. This could open the door for novel and exciting applications for small rotorcraft.

Appendix

Table 4 Unmanned rotorcraft dataset.

Name	Rotor	Prop.	W, kg	D, m	W_p, kg	m_b, kg	E_b, Wh	t, min	S, km	H, km	V, km h^{-1}
Acecore NEO X8 [32]	O8	el.	19.0	0.46	9.00	–	511	25	16.0	–	91
Acecore NOA 6 [32]	O6	el.	37.0	0.71	20.0	–	1100	55	16.0	–	85
Acecore ZOE X4 [32]	O4	el.	12.0	0.46	6.00	–	377	35	16.0	–	91
Aerialtronics Altura Zenith [43]	C4	el.	9.65	0.41	3.00	–	444	40	–	–	72
Aerovironment Vapor 55 mx [40]	O1	el.	29.5	2.29	10.0	–	–	75	32.0	3.7	54
Airbus VSR700 [19]	O1	th.	700	7.20	–	–	–	480	–	6.0	185
Alpha ALPHA 800 [26]	O1	th.	14.0	1.80	3.00	–	–	180	50.0	3.0	–
Autel EVO 2 [37]	O4	el.	2.00	0.23	–	0.36	82.0	40	25.0	7.0	72
Autel EVO 2 DUAL 640T [37]	O4	el.	2.00	0.23	0.85	0.36	82.0	38	22.0	7.0	72
Autel EVO 2 Enterprise v3 [37]	O4	el.	–	0.23	–	–	–	42	–	7.0	72
Autel EVO 2 Pro V3 [37]	O4	el.	2.00	0.23	–	0.36	82.0	40	25.0	7.0	72
Autel EVO 2 RTK [37]	O4	el.	–	0.23	–	–	–	36	21.0	7.0	72
Autel EVO lite [37]	O4	el.	0.83	0.22	–	–	68.7	40	24.0	4.0	65
Autel EVO nano [37]	O4	el.	0.25	0.12	–	–	17.3	28	16.8	4.0	54
Autoflight F240 Hot Wheels [47]	O8	el.	320	–	100	–	–	–	–	–	–
Bitcraze Crazyflie 2.1 [35]	O4	el.	0.04	0.04	0.01	0.01	0.74	7	1.0	–	–
BlackSwift E2 [33]	O4	el.	8.80	0.47	2.00	–	944	60	–	–	–
Boeing Cargo Air Vehicle [48]	C6	el.	545	–	225	–	–	–	–	–	–
Challenger CP-A5 [27]	O6	el.	18.0	0.71	7.00	–	–	75	5.0	4.5	33

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Table 4 – continued from previous page

Name	Rotor	Prop.	W, kg	D_R, m	W_p, kg	m_b, kg	E_b, Wh	t, min	S, km	H, km	V, km h^{-1}
Challenger CPXD-6R-16L [27]	O6	el.	37.5	–	16.0	–	–	21	3.0	2.0	36
Challenger H-4 [27]	O4	el.	21.5	–	11.0	–	–	60	1.0	4.5	65
Challenger H-4 Plus [27]	C4	el.	32.0	–	18.0	–	–	60	1.0	4.5	65
Challenger H-6 [27]	O6	el.	30.0	–	18.0	–	–	75	1.0	4.5	72
Challenger H-8 [27]	O8	el.	52.5	–	36.0	–	–	120	1.0	4.5	86
Challenger ODIN-X [27]	O1	th.	44.5	2.41	20.0	–	–	120	–	3.2	–
Challenger ODIN-XL [27]	O1	th.	85.0	2.70	30.0	–	–	120	–	3.2	–
DJI Agras T10 [18]	O4	el.	24.8	0.84	8.00	3.80	492	17	–	4.5	25
DJI Agras T16 [18]	O4	el.	42.0	0.84	16.0	–	907	18	–	2.0	25
DJI Agras T20 [18]	O4	el.	47.5	0.84	20.0	6.40	932	15	–	2.0	25
DJI Agras T30 [18]	O6	el.	78.0	0.97	41.5	10.1	1500	20	–	4.5	25
DJI Agras T40 [18]	C4	el.	101	1.37	51.0	12.0	1570	18	–	–	–
DJI Air 2s [18]	O4	el.	0.59	0.18	–	0.20	41.4	30	18.5	5.0	68
DJI Avata [18]	D4	el.	0.41	0.07	–	0.16	35.7	18	11.6	5.0	97
DJI FPV [18]	O4	el.	0.80	0.13	–	0.29	44.4	20	16.8	6.0	97
DJI Inspire 2 [18]	O4	el.	4.25	0.38	0.81	0.52	97.6	27	–	5.0	94
DJI Matrice 30 [18]	O4	el.	4.00	0.41	–	1.37	263	41	–	5.0	83
DJI Matrice 300 RTK [18]	O4	el.	9.00	0.53	2.70	2.70	548	55	–	5.0	83
DJI Matrice 600 Pro [18]	O6	el.	15.5	0.53	6.00	4.08	780	32	–	4.5	65
DJI Mavic 2 Enterprise [18]	O4	el.	1.10	0.22	0.19	0.30	59.3	31	–	6.0	72
DJI Mavic 3 [18]	O4	el.	0.90	0.24	–	0.34	77.0	46	30.0	6.0	68
DJI Mavic 3 Classic [18]	O4	el.	0.90	0.24	–	0.34	77.0	40	30.0	6.0	76
DJI Mavic 3 Enterprise [18]	O4	el.	1.05	0.24	–	0.34	77.0	45	32.0	6.0	54
DJI Mavic Air 2 [18]	O4	el.	0.57	0.18	–	0.20	40.4	34	18.5	5.0	68
DJI Mini 2 [18]	O4	el.	0.25	0.12	–	0.09	17.3	31	–	4.0	58
DJI Mini 3 Pro [18]	O4	el.	0.25	0.15	–	0.12	28.4	47	25.0	4.0	58
DJI Mini SE [18]	O4	el.	0.25	–	–	0.08	17.3	30	–	3.0	47
DJI P4 Multispectral [18]	O4	el.	1.49	0.24	–	0.47	89.2	27	–	6.0	58
DJI Phantom 4 PRO [18]	O4	el.	1.39	0.24	–	0.47	89.2	30	–	6.0	72
DJI Phantom 4 PRO v2.0 [18]	O4	el.	1.38	0.24	–	0.47	89.2	40	–	6.0	72
DJI Phantom 4 RTK [18]	O4	el.	1.39	0.24	–	0.47	89.2	30	–	6.0	58
DJI Spark [18]	O4	el.	0.30	0.12	–	0.10	16.9	16	–	4.0	50
Draganfly Commander [42]	O4	el.	3.75	–	1.00	–	–	25	–	–	50
Draganfly Commander 3 XL [42]	O4	el.	25.0	–	11.8	–	–	50	38.6	–	72
Draganfly Heavy Lift Drone [42]	O8	el.	74.0	0.82	30.0	–	4560	54	20.0	–	79
Draganfly X4P [42]	O4	el.	3.75	–	1.00	–	–	18	–	2.4	50
Ehang 216L [49]	C8	el.	600	–	220	–	–	21	35.0	–	130
FLIR Black Hornet [46]	O1	el.	0.03	0.12	–	–	–	25	–	–	22

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Table 4 – continued from previous page

Name	Rotor	Prop.	W, kg	D_R, m	W_p, kg	m_b, kg	E_b, Wh	t, min	S, km	H, km	V, km h^{-1}
FLIR Skyranger R70 [46]	O4	el.	5.00	–	3.50	–	99.0	60	–	–	50
Flyability Elios 2 [45]	D4	el.	1.45	0.13	–	0.55	98.8	10	–	–	23
Flyability Elios 3 [45]	D4	el.	2.50	0.13	0.60	0.62	99.0	12	–	–	–
Freefly Alta X [44]	O4	el.	34.9	0.84	15.3	–	710	50	–	–	–
Freefly Astro [44]	O4	el.	6.67	0.53	1.50	1.03	157	38	–	–	–
Inspired Flight IF1000 [30]	O6	el.	9.90	0.43	3.30	–	–	35	10.0	5.4	65
Inspired Flight IF1200 [30]	O6	el.	20.0	0.46	8.00	5.80	977	33	–	6.8	79
Inspired Flight IF750 [30]	O4	el.	6.20	0.43	2.00	–	–	30	10.0	5.9	65
Leonardo AWHERO [20]	O1	th.	200	4.00	85.0	–	–	360	–	4.3	167
Parrot Anafi Ai [36]	O4	el.	0.90	–	–	0.37	29.9	32	22.5	5.0	61
Riegl RiCOPTER [29]	C4	el.	25.0	–	6.50	–	–	30	14.5	4.0	50
Riegl RiCOPTER-M [29]	C4	el.	40.0	–	15.0	–	–	25	12.5	3.0	50
Ryze Tello Edu [34]	O4	el.	0.08	0.08	–	0.03	4.18	13	–	0.0	29
Schiebel CAMCOPTER S-100 [25]	O1	th.	200	3.40	50.0	–	–	360	200.0	5.5	240
SkyFront PERIMETER 4 [28]	O4	th.	15.0	–	4.00	–	–	300	50.0	–	57
SkyFront PERIMETER 8 [28]	O8	th.	19.0	–	5.50	–	–	300	100.0	–	–
SkyFront TAILWIND [28]	O4	th.	12.0	–	3.00	–	–	300	250.0	0.1	56
Skydio 2+ Enterprise [39]	O4	el.	0.80	0.17	–	–	48.8	27	6.0	4.5	58
Skydio X2E [39]	O4	el.	0.80	0.17	–	–	–	35	6.0	3.6	40
Steadicopter Black Eagle 50 [23]	O1	th.	35.0	2.16	5.00	–	–	240	–	3.0	126
Steadicopter Black Eagle 50H [23]	O1	th.	50.0	2.80	12.0	–	–	210	–	5.0	126
Threod Systems TITAN [31]	O4	el.	17.0	1.05	6.00	–	–	50	5.0	–	–
UAVOS UVH 170 [22]	O1	th.	45.0	2.60	10.0	–	–	300	350.0	2.5	120
UAVOS UVH 25EL [22]	O1	el.	25.0	2.60	5.00	–	–	90	67.0	3.5	100
UAVOS UVH 500 [22]	O1	th.	500	6.28	160	–	–	420	840.0	2.5	209
UAVOS UVH R22 [22]	O1	th.	635	7.70	180	–	–	360	1020.0	4.2	189
UAVOS UVM 2E [22]	O4	el.	21.0	0.76	1.00	–	–	55	10.0	1.6	55
UMS Skeldar V-150 [24]	O1	th.	150	3.50	30.0	–	–	240	100.0	3.0	120
UMS Skeldar V-200 [24]	O1	th.	235	4.60	–	–	–	300	200.0	3.0	150
Velos Rotors UAV [21]	O1	el.	24.0	1.95	10.0	–	–	75	–	–	130
Velos Rotors V3 [21]	O1	el.	25.0	1.95	10.0	–	–	90	–	–	130
Volocopter VoloDrone [41]	U16	el.	800	–	200	–	–	30	40.0	–	110
Yuneec H3 [38]	O6	el.	2.00	–	–	–	79.8	25	–	0.5	72
Yuneec H520E [38]	O6	el.	1.65	–	–	–	79.8	28	–	0.5	61
Yuneec H850 RTK [38]	O6	el.	8.80	–	3.00	–	584	65	–	5.0	72
Yuneec Typhoon H Plus [38]	O6	el.	2.00	–	–	–	79.8	28	–	0.5	48

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